Week 12

MICROECONOMIC THEORY
ECON 323 502/503

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We have covered the analysis of

1. competitive market (many price-taking firms)
2. monopoly market (one firm who effects the price)

A market structure with a few firms is called an oligopoly market.

- E.g., wireless carriers in the US

In an oligopoly market, a firm's output/pricing decision affects other firms.

Eventually, we want to know how does the output/price/welfare compare with competitive/monopoly market?

We will do this in two steps.

- First, we will study a branch of economics and mathematics that is concerned with finding reasonable predictions in strategic situations. This is known as "Game Theory."

- Then, we will apply game theory to our oligopoly model.

Chapter 13. Game Theory and Competitive Strategy

Related Readings in the textbook

- 13.1 Gaming and Strategic Decisions
- 13.2 Dominant Strategies
- 13.3 The Nash Equilibrium Revisited
- 13.5 Sequential Games
- 12.4 Competition versus Collusion: The Prisoners' Dilemma

Game: Situation in which players make strategic decisions that take into account each other's actions and and responses.

We focus on **noncooperative game**: Game in which negotiation and enforcement of binding contracts are not possible.

Three components of a game:

players There are at least two agents.

payoff Value associated with a possible outcome.

strategy Rule or plan of action for playing a game.

Example of a Game:

Split or Steal Game https://www.facebook.com/classictvmoments/videos/100000-steal/2294216330611721/

Can you identify the players, payoffs, and strategy of this game?

- Two **players**, the man and the woman.
- Each player has two **strategies**: to split and to steal.
- **Payoffs**:
  - If both choose to split, then each player wins £50,075;
  - If only one chooses to steal, then the “steal”-player wins £100,150 and the other receives £0;
  - If both choose to steal, then each player receives £0.

How to represent a game?

Payoff matrix: Table showing payoff to each player given its decision and the decision of the other player.

Row Player

Column Player

Two potential actions of the Column Player

		Woman	
		Split	Steal
Man	Split	£50,075, £50,075	£0, £100,150
	Steal	£100,150, £0	£0, £0

Payoff Matrix of the Split-or-Steal Game

Two potential actions of the Row Player

Payoff of the Row Player

Payoff of the Column Player

Example: Rock, Paper, Scissors

- Two player, simultaneous game
- Rock>Scissors>Paper>Rock
- Winner's payoff 1; Loser's payoff -1; Tie → zero payoff

		Player 2		
		Rock	Paper	Scissors
Player 1	Rock			
	Paper			
	Scissors			

Example: Rock, Paper, Scissors

- Two player, simultaneous game
- Rock>Scissors>Paper>Rock
- Winner's payoff 1; Loser's payoff -1; Tie → zero payoff

		Player 2		
		Rock	Paper	Scissors
Player 1	Rock	0, 0	-1, 1	1, -1
	Paper	1, -1	0, 0	-1, 1
	Scissors	-1, 1	1, -1	0, 0

Example: Battle of the Sexes

- A girl (who likes movies) and a boy (who likes football).
- Simultaneously and independently decide whether to go for a movie or go to a football game.
- Happier by being together and enjoying his/her hobby
- No utility when missing each other

		Boy	
		Movie	Football
Girl	Movie	3, 2	0, 0
	Football	0, 0	2, 3

Players: Girl, Boy
Strategy of Girl/Boy: Movie, Football. Each has 2 strategies
Strategy Profile: (Strategy of Girl, Strategy of Boy). 4 strategy profiles in total.
Payoff Profile: (Payoff of Girl, Payoff of Boy)

Example: Prisoners' Dilemma

https://www.youtube.com/watch?v=t9Lo2fgxWHw

Example: Prisoners' Dilemma

- Two suspects are arrested by the police because of two charges, a minor charge (e.g., loitering) and a major one (robbery). They are interrogated in separate rooms.

- The policemen have evidence to prove the minor charge, but lack evidence to prove the major charge. They can only prove the major charge if they can convince at least one of the suspects to betray his accomplice and testify to the crime.

- Each suspect has two actions to take: to confess robbery or to deny robbery.

Example: Prisoners' Dilemma

		Prisoner 2	
		Confess	Deny
Prisoner 1	Confess	-5, -5	0, -10
	Deny	-10, 0	-1, -1

Dominant Strategy

- A **Dominant Strategy** is a strategy that is optimal (including the special case of being one of the multiple optimal strategies) no matter what an opponent does.

Is there a dominant strategy for the man/woman in the split-or-steal game?

		Woman	
		Split	Steal
Man	Split	£50,075, £50,075	£0, £100,150
	Steal	£100,150, £0	£0, £0

For the man, playing steal is the dominant strategy, because no matter what the woman does, stealing is always optimal.

Similarly, for the woman, playing steal is also the dominant strategy.

Is there a dominant strategy for the girl/boy in the Battle of the Sexes game?

		Boy	
		Movie	Football
Girl	Movie	3, 2	0, 0
	Football	0, 0	2, 3

No dominant strategy for the girl. When the boy goes for movie, it is optimal for the girl to go for movie; but when the boy goes for football, it is optimal for the girl to go for football.

Similarly, no dominant strategy for the boy.

Is there a dominant strategy for either Player in the Rock-Paper-Scissors game?

		Player 2		
		Rock	Paper	Scissors
Player 1	Rock	0, 0	-1, 1	1, -1
	Paper	1, -1	0, 0	-1, 1
	Scissors	-1, 1	1, -1	0, 0

No dominant strategy for Player 1. The best strategy of Player 1 changes when Player 2's strategy changes.

Similarly, no dominant strategy for Player 2.

Is there a dominant strategy for the Prisoners' Dilemma game?

		Prisoner 2	
		Confess	Deny
Prisoner 1	Confess	-5, -5	0, -10
	Deny	-10, 0	-1, -1

Prisoner 1: Confess is a dominant strategy.

Prisoner 2: Confess is a dominant strategy.